

SIG Math

Backpropagation & Problems of PINNs



Backpropagation

- Essentially calculating the direction of increase in Loss (the gradient)
- Use this information, we going against the gradient to decrease the loss

$$\mathcal{L}(\mathbf{u}_{pred}, \mathbf{u}_{true}) = \frac{1}{2} \sum_{j=1} (\mathbf{u}_{pred}^j - \mathbf{u}_{true}^j)$$

$$\mathcal{L}(\mathbf{u}_{pred}, \mathbf{u}_{true}) = \frac{1}{2} \sum_{j=1} (f(\mathbf{w} \cdot \mathbf{x} + b)^j - \mathbf{u}_{true}^j)$$

Derivatives

$$\mathcal{L}(\mathbf{u}_{pred}, \mathbf{u}_{true}) = \frac{1}{2} \sum_{j=1} (f(\mathbf{w} \cdot \mathbf{x} + b)^j - \mathbf{u}_{true}^j)$$

$$\frac{\partial \mathcal{L}}{\partial w_j} = \frac{\partial \mathcal{L}}{\partial a} \frac{\partial a}{\partial z} \frac{\partial z}{\partial w_j}$$

$$z = \mathbf{w} \cdot \mathbf{x} + b$$

$$a = f(z)$$

Caveat in vanishing gradient generally

Note that with the chain of derivatives, if any of them goes to 0, there will be zero gradient. This is the [vanishing gradient problem](#)

Gradient descent

$$\mathbf{w}_{new} = \mathbf{w}_{old} - \alpha \nabla_{\mathbf{w}} \mathcal{L}$$

Attendance

go.uic.edu/coeorgtracker



Troubles of PINNs

- Imbalance between loss terms
- It can not handle sharp gradients or singularities well
- Poor scaling to high-dimensional PDEs

Extra readings on bettering PINNs

- Attention mechanism in Neural Networks and PINNs:
 - [Equity-premium prediction: Attention is all you need](#) (for more general overview of attention mechanism)
 - [Improved physics-informed neural network in mitigating gradient related failures](#)
- Localized viscosity for handling sharp gradients
 - [Physics-Informed Neural Networks with Adaptive Localized Artificial Viscosity](#)
- (Advanced) Kolmogorov-Arnold networks
 - [KAN: Kolmogorov-Arnold Networks](#)
 - [Physics-informed neural networks with hybrid Kolmogorov-Arnold network and augmented Lagrangian function for solving partial differential equations.](#)